

Curriculum Vitae | Laura Ann Busch

Summary of main achievements

- **Scientific achievements:** Established low $^{12}\text{C}/^{13}\text{C}$ ratios in seven low-mass protostars and used spatially resolved observations of a high-mass hot core to investigate molecular formation and destruction pathways.
- **Observational expertise:** Developed expertise spanning single-dish and interferometric observations, molecular spectral-line and isotopologue analysis, and LTE/non-LTE modelling, from observing proposals and data reduction to scientific interpretation.
- **Independence and collaboration:** Successfully led observing proposals and contributed to major international collaborations and surveys, including PRODIGE and ACES. Regularly presented research at international conferences, workshops and seminars, building a broad interdisciplinary network in astrochemistry and star formation.

Current position

since 08/01/2024 **Postdoctoral Position**, *Max-Planck-Institut für extraterrestrische Physik*, Garching bei München, Germany.
Advisors: Dr. Jaime Pineda and Prof. Dr. Paola Caselli.
Details: Involved in the PRODIGE and ACES collaboration, analysing observational data.

Education

01/03/2020 – 14/12/2023 **Ph.D Astrophysics/Astronomy**, *Rheinische Friedrich-Wilhelms-Universität*, Bonn, Germany.
Ph.D thesis title: Formation and desorption of organic molecules: Resolving their emission in a prominent protostellar hot core and its outflow.
Affiliation: Max-Planck-Institut für Radioastronomie, Bonn.
Advisors: Dr. Arnaud Belloche and Prof. Dr. Karl Menten.

01/04/2017 – 31/01/2020 **MSc Astrophysics**, *Rheinische Friedrich-Wilhelms-Universität*, Bonn, Germany.
MSc thesis title: The peculiar ISM at the edge of the Central Molecular Zone.
Date: 01/02/2019 – 31/01/2020.
Affiliation: Max-Planck-Institut für Radioastronomie, Bonn.
Advisors: Dr. Denise Riquelme and Prof. Dr. Karl Menten.

01/10/2013 – 24/03/2017 **BSc Physics**, *Rheinische Friedrich-Wilhelms-Universität*, Bonn, Germany.
BSc thesis title: The obscuring torus of the elliptical galaxy NGC 1052.
Date: 10/2016 – 03/2017 (4 months).
Affiliation: Max-Planck-Institut für Radioastronomie, Bonn.
Advisor: Prof. Dr. Eduardo Ros.

Additional experience

Student employee, Max-Planck-Institut für Radioastronomie, Bonn, Germany,

Date: 16/06/2018 – 27/07/2018 and 04/10/2018 – 31/01/2019

Description: Project leading to first publication entitled '*The dynamically young outflow of the Class 0 protostar Cha-MMS1*', supervised by Dr. Arnaud Belloche

Teaching, Argelander-Institut für Astronomie, Bonn, Germany

Description: Tutorials and exam preparations, supervision, and evaluation

Dates & Hours: Summer terms 2018 & 2021; Winter terms 2017/18 & 2018/19; ~20 h/month

Observation training, Remote and on site at the IRAM 30m (Spain), Effelsberg 100m (Germany), and APEX 12m (Chile) telescopes

Journal Referee for Astronomy & Astrophysics and Astrophysical Journals, since 2024

Skills

Programming: Python (NumPy, SciPy, Matplotlib, Astropy, SpectralCube), LaTeX

Astronomical Software: GILDAS (CLASS, Weeds, MAPPING, GREG, CLIC), MADCUBA, Radex, CARTA

Languages: German (native), English (fluent), French (intermediate)

Memberships & collaborations

since 01/2024

ALMA CMZ Exploration Survey (ACES), Max-Planck-Institut für extraterrestrische Physik, Garching bei München, Germany.

Description: International Collaboration studying star formation in the Central Molecular Zone (CMZ) in the Galactic centre region; data was observed with the Atacama Large (sub)mm Array (ALMA).

Role: Analysis of ACES data.

since 01/2024

Protostars and Disks: Global Evolution" (PRODIGE), Max-Planck-Institut für extraterrestrische Physik, Garching bei München, Germany.

Description: Collaboration between the Max Planck Society and Institut de radioastronomie millimétrique (IRAM) studying kinematic and chemistry in the Perseus and Taurus low-mass star-forming regions.

Role: Reduction and analysis of PRODIGE data.

01/03/2020 –
31/12/2022

Collaborative Research Centre (CRC) 956, Max-Planck-Institut für Radioastronomie, Bonn, Germany.

Description: Main financial support during Ph.D.

01/03/2020 –
14/12/2023

International Max Planck Research School (IMPRS) for Astronomy and Astrophysics, Max-Planck-Institut für Radioastronomie, Bonn, Germany.

Description: Additional financial support during Ph.D.

01/04/2017 –
14/12/2023

Bonn-Cologne Graduate School (BCGS) for Physics and Astronomy, Rheinische Friedrich-Wilhelms-Universität, Bonn, Germany.

Description: Additional support during MSc and Ph.D.

List of Publications

submitted	L. A. Busch , K. M. Dutkowska, L. Colzi, I. Jiménez-Serra et al., <i>Hunting hot cores with the ACES: A new star-forming cluster found in Sagittarius B2</i> , submitted to A&A.
2026	S. N. Longmore, J. Bally, A. T. Barnes et al. (incl. L. A. Busch), <i>ACES – I: Overview</i> , arXiv:2602.20340 .
2026	A. Ginsburg, D. L. Walker, A. T. Barnes et al. (incl. L. A. Busch), <i>ACES – II: 3 mm continuum images</i> , arXiv:2602.20240 .
2026	D. L. Walker, A. Ginsburg, A. T. Barnes et al. (incl. L. A. Busch), <i>ACES – III: Molecular line data reduction and HNCO and HCO⁺ data</i> , arXiv:2602.20276 .
2026	X. Lu, D. L. Walker, A. Ginsburg et al. (incl. L. A. Busch), <i>ACES – IV: Data of the two intermediate-width spectral windows</i> , arXiv:2602.20445 .
2026	P.-Y. Hsieh, D. L. Walker, A. Ginsburg et al. (incl. L. A. Busch), <i>ACES – V: CS(2–1), SO(2₃–1₂), CH₃CHO(5_{1,4}–4_{1,3}), HC₃N(11–10), and H40α lines data</i> , arXiv:2603.00863 .
30/04/2026	M. T. Valdivia-Mena, J. E. Pineda, C. Gieser, P. Caselli, D. M. Segura-Cox, Y. Lin, M. J. Maureira, T.-H. Hsieh, L. A. Busch , <i>PRODIGE – Envelope to disk with NOEMA– VIII. Sulfur oxides trace a shock caused by a streamer in the inner envelope of a protostar</i> , A&A, 709, A53, 12pp .
21/04/2026	L. A. Busch , J. E. Pineda, D. M. Segura-Cox, P. Caselli et al., <i>PRODIGE – envelope to disk with NOEMA – VII. (Complex) organic molecules in the NGC1333 IRAS 4B1 outflow: A new laboratory for shock chemistry</i> , A&A, 708, A315, 13pp .
05/11/2025	K. M. Dutkowska, G. Vermariën, S. Viti, I. Jiménez-Serra, L. Colzi, L. A. Busch et al., <i>Chemical templates of the Central Molecular Zone – Shock and protostellar object signatures under Galactic Center conditions</i> , A&A, 703, A46, 22pp .
12/09/2025	C. Gieser, P. Caselli, D. M. Segura-Cox, J. E. Pineda, L. A. Busch et al., <i>ANTIHEROES-PRODIGE: Quantifying the connection from the envelope to the disk with the IRAM 30 m telescope and NOEMA – I. Infalling streamers in L1448N</i> , A&A, 701, A165, 20pp .
22/08/2025	J. J. Miranzo-Pastor, A. Fuente, D. Navarro-Almaida, J. E. Pineda, D. M. Segura-Cox, P. Caselli, R. Martin-Domenech, M. T. Valdivia-Mena, T. Henning, T.-H. Hsieh, L. A. Busch , et al., <i>PRODIGE – envelope to disk with NOEMA – VI. The missing sulfur problem</i> , A&A, 700, A251, 30pp .
21/07/2025	L. A. Busch , J. E. Pineda, O. Sipilä, D. M. Segura-Cox, P. Caselli et al., <i>PRODIGE – envelope to disk with NOEMA – V. Low ¹²C/¹³C ratios for CH₃OH and CH₃CN in hot corinos</i> , A&A, 699, A359, 27pp .
31/03/2025	T.-H. Hsieh, J. E. Pineda, D. M. Segura-Cox, P. Caselli, M. J. Maureira, L. A. Busch et al., <i>Resolved gas temperatures and ¹²C/¹³C ratios in SVS13A from ALMA Observations of CH₃CN and CH₃¹³CN</i> , A&A, 696, A1, 14pp .
02/12/2024	C. Gieser, J. E. Pineda, D. M. Segura-Cox, P. Caselli, M. T. Valdivia-Mena, M. J. Maureira, T.-H. Hsieh, L. A. Busch et al., <i>PRODIGE – envelope to disk with NOEMA – IV. An infalling gas bridge surrounding two Class 0/I systems in L1448N</i> , A&A, 692, A55, 16pp .

24/01/2024	L. A. Busch , A. Belloche, R. T. Garrod et al., <i>Shocking Sgr B2 (N1) with its own outflow: A new perspective on segregation between O- and N-bearing molecules</i> , A&A , 681, A104, 52pp.
21/12/2022	L. A. Busch , D. Riquelme, R. Güsten et al., <i>Living on the edge of the Milky Way's central molecular zone: G1.3 is the likely candidate for gas accretion into the CMZ</i> , A&A , 668, A183, 23pp.
16/09/2022	L. A. Busch , A. Belloche, R. T. Garrod et al., <i>Resolving desorption of complex organic molecules in a hot core: Transition from non-thermal to thermal desorption or two-step thermal desorption?</i> , A&A , 665, A96, 73pp.
22/01/2020	L. A. Busch , A. Belloche, S. Cabrit et al., <i>The dynamically young outflow of the Class 0 protostar Cha-MMS1</i> , A&A , 633, A126, 16pp.

Accepted observing PI proposals

‘*Hot, hotter, gas at the edge of the CMZ?*’, Effelsberg 100m, data acquired in winter 2023 (16 h)
‘*Shocking! Complex organic molecules at the edge of the CMZ*’, APEX 12m, data acquired in spring 2023 (57/36 h)
‘*Probing molecular complexity in a new laboratory: Hot-corino & shock chemistry towards IRAS 4B1 and its outflow*’, NOEMA+30m, NOEMA data acquired in summer 2026 (~25 h), not yet delivered, complementing IRAM 30m observations (29 h) not yet scheduled

Presentations at conferences & seminars

02–09/08/2026	Contributed Talk , <i>46th COSPAR Scientific Assembly</i> , Florence, Italy. Title: Molecular complexity in extreme environments: Chemistry in the Central Molecular Zone with ACES.
02–09/08/2026	Contributed Poster , <i>46th COSPAR Scientific Assembly</i> , Florence, Italy. Title: Probing molecular complexity in Class 0 protostars with PRODIGE.
25–29/08/2024	Contributed Poster , <i>Interstellar Disparity: N- vs O-bearing molecules</i> , Leiden, Netherlands. Title: New clues on O/N segregation from Sgr B2(N1) and its outflow.
19–22/08/2025	Contributed Talk , 839. <i>WE-Heraeus-Seminar: Charting the cosmos: From cosmic stellar nurseries to evolved stars using high powered telescopes</i> , Ingelheim, Germany. Title: From clouds to stars: What $^{12}\text{C}/^{13}\text{C}$ ratios reveal about molecular inheritance.
04–07/11/2024	Contributed Talk , <i>Fractionation of isotopes in space II – from the Solar System to galaxies</i> , Florence, Italy. Title: Do we observe ^{13}C enrichment towards young low-mass protostars?.
01–05/07/2024	Contributed Talk , <i>The European Astronomical Society Annual Meeting</i> , Padova, Italy. Title: Shocking Sgr B2(N1) with its own outflow - A new perspective on O/N segregation in molecules.
28/11–01/12/2022	Contributed Talk , <i>Connecting the hot core and hot corino phenomena II</i> , Bordeaux, France. Title: Resolving emission of complex organic molecules in the hot core Sgr B2(N1) and its outflow with ReMoCA.

- 03–07/10/2022** **Contributed Talk**, *From Clouds to Planets II: The Astrochemical Link*, Berlin, Germany.
Title: Exposing the COMfort zone of Sgr B2(N1) by resolving the desorption of Complex Organic Molecules in the hot core.
- 21/09/2022** **Solicited Talk**, *Journal club at the Max Planck Institute for Extraterrestrial Physics*, virtual.
Title: Resolving desorption of complex organic molecules in a hot core. Transition from non-thermal to thermal desorption or two-step thermal desorption? (A&A, 665, A96).
- 27/06–01/07/2022** **Contributed Talk**, *The European Astronomical Society Annual Meeting*, Valencia, Spain.
Title: Exposing the COMfort zone of SgrB2(N1) by resolving the desorption of Complex Organic Molecules in the hot core.
- 02–04/03/2022** **Contributed Talk**, *Meeting of ALMA Young Astronomers*, virtual.
Title: Resolving the COMfort zone of Sgr B2 (N1) with ReMoCA.
- 02–05/02/2020** **Contributed Talk**, *Science with the Atacama Pathfinder EXperiment (APEX)*, Ringberg, Germany.
Title: High- and low-velocity gas at the edge of the Central Molecular Zone.

Workshops & summer schools

- 06–10/07/2026** **Workshop**, *Team meeting of ACES collaboration*, Manchester, United Kingdom.
- 23–27/03/2026** **Workshop**, *Hands-on training for astrochemical modelling with UCLCHEM*, Bonn, Germany.
- 20–24/10/2025** **Workshop**, *Astrochemistry in the Broadband Era: ngVLA & ALMA WSU*, Portland, USA.
- 10–11/06/2025** **Workshop**, *Team meeting of PRODIGE collaboration*, Heidelberg, Germany.
- 10–14/06/2024** **Workshop**, *Kick-off meeting of ACES collaboration*, Garching bei München, Germany.
- 21–25/11/2022** **Summer school**, *11th IRAM Interferometry School*, Grenoble, France.
- 08–28/04/2022** **Workshop**, *Principles and tools of statistics, data analysis and machine learning: I. Statistics*, Bonn, Germany.
- 12–23/07/2021** **Summer school**, *International Summer School on the Interstellar Medium of Galaxies, from the Epoch of Reionisation to the Milky Way*, virtual.

Extra curricular activities

- 05/04/2023** **Public talks at Bad Münstereifel**, *Interstellare komplexe Moleküle und wo sie zu finden sind*, Bad Münstereifel, Germany.
Description: Talk series in German for the interested public audience organised by Max-Planck-Institut für Radioastronomie, Bonn.
- 27/03/2023** **Montagsvorträge in der Sternwarte**, *Interstellare komplexe Moleküle und wo sie zu finden sind*, Bonn, Germany.
Description: Talk series in German for the interested public audience at the old Bonn observatory.

07/2021 & 01/2023	Local Organising Committee , <i>for IMPRS student seminar</i> , Bonn, Germany. <i>Description:</i> Plan, organise, and host the Ph.D student seminar.
27/03/2023	Highlight der Physik , <i>Max-Planck-Institut für Radioastronomie</i> , Bonn, Germany. <i>Description:</i> Representing the research departments of MPIfR at the exposition during this event, which was initiated to promote research in physics and astronomy amongst the public.
2017,2018,2023, 2024	Girl's Day , <i>MPI für Radioastronomie & für extraterrestrische Physik</i> , Bonn & Garching bei München, Germany. <i>Description:</i> Organise and host workshops to promote science amongst female school students.
